

Kinematics-Grounded Motion Planning for Robots

Precoded trajectories



Yaskawa Inc.



Obstacles were known, static and path was guaranteed to be feasible.

Motion planning is everywhere



Image source: [Kinova's Jaco page](#)



Image source: [Springer article](#)



Image source: [Research article](#)

Robots have moved from industrial settings into everyday life.

Assistive and collaborative robots now interact directly with people.

Motion planning with guarantees is essential for safe human–robot interaction.

Demonstration



To ease transfer, one can embody the robot.
EPFL-LASA

Can the robot reproduce safely the path even when starting with different initial conditions?

The learned control policy must be safe.

What would you check?

Convergence

Stability at target

Bounded forces

Topics covered in MICRO-462 Learning and Adaptive Control for Robots
(Spring 2027)

When motion fails



Can the robot execute the **whole path**?

Even if the goal is reachable, this does guarantee that there is a feasible path to reach it.

What would you check?

Continuous path in joint space

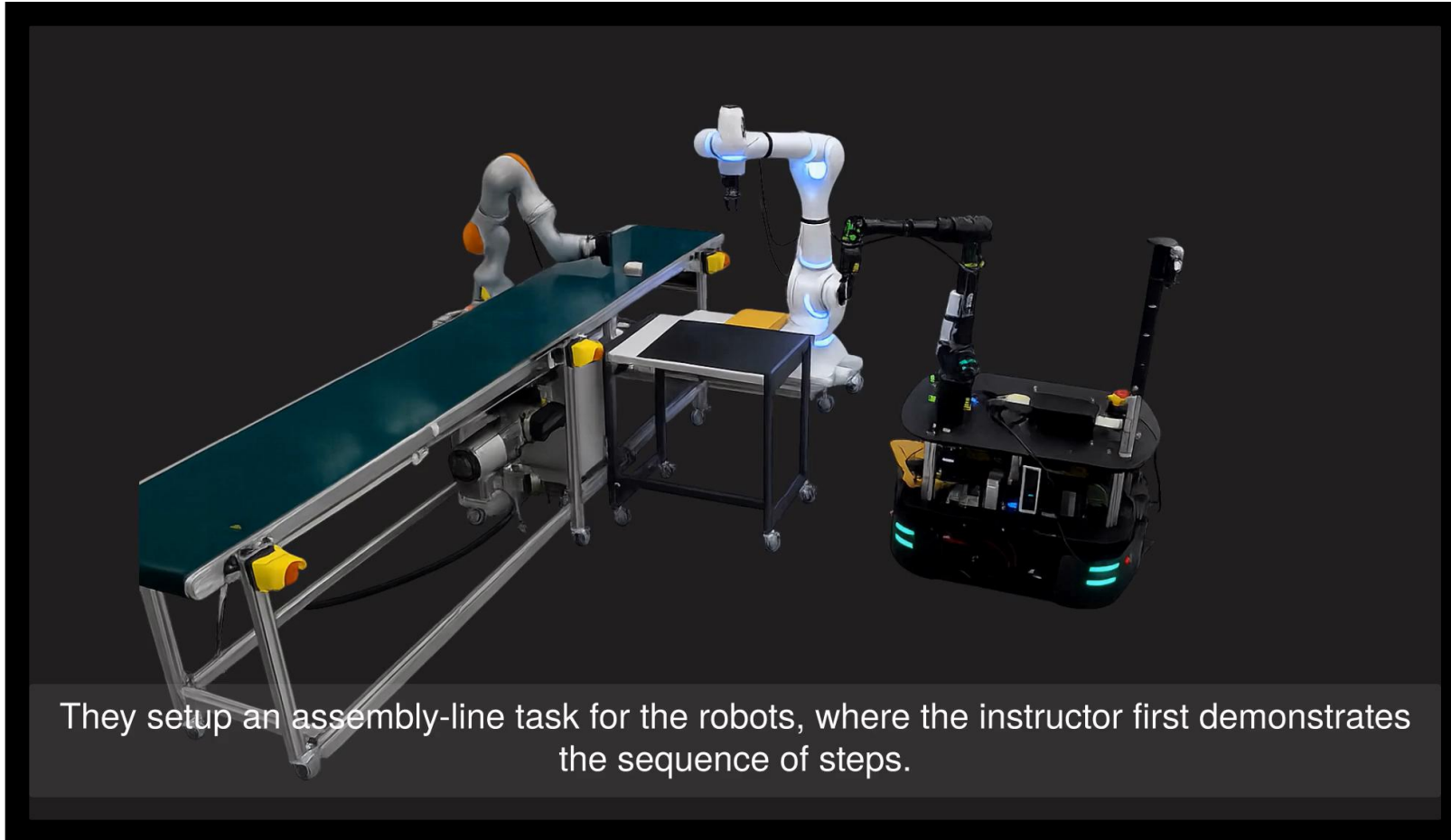
Joint limits and collisions

Lost motion directions (singularities)

Real world consequences of failed robot planning.

Video source: [Achille Verheye on Medium](#)

Transfer Knowledge across robots



If the robots differ in their number of links or joint configurations, a path that was feasible for one robot may become infeasible for another robot. Need methods to transfer easily across different kinematics.

Global analysis and guarantees

Feasibility over the entire motion.

With explicit assumptions.

01 The allowed robot postures

Where do joint limits, collisions and singularities restrict motion?

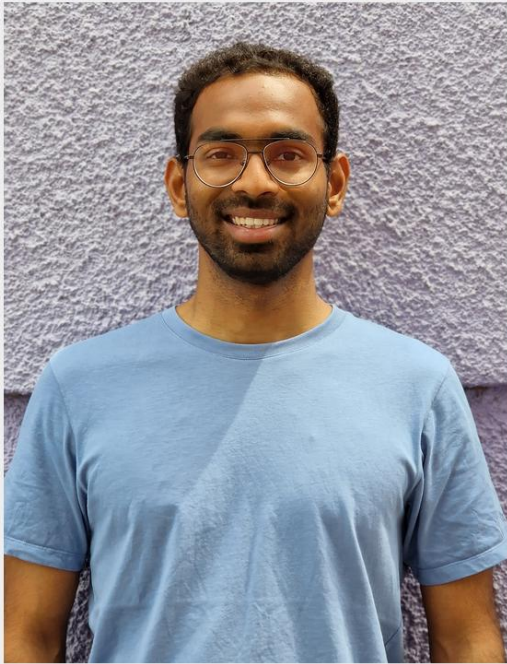
02 The connections between postures

Which postures can the robot move between continuously?

03 The evidence for execution

Which constraints hold along the path, and what remains unverified?

A guarantee is relative to the model and constraints. Checking isolated samples alone does not certify the motion between them.



Durgesh Salunkhe

Postdoctoral researcher
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Course ENG-654



Doctorate in robot motion

French National Centre for
Scientific Research (CNRS)
Laboratory of Digital Sciences
of Nantes (LS2N)
Nantes, France



Adapting skills across robots

Learning Algorithms and
Systems Laboratory (LASA)
EPFL, Lausanne, Switzerland

8

years in robotics

Robot design, optimisation & motion planning

Science Robotics

Demonstrate once, execute on many
Co-first author, 2026



The International Journal
of Robotics Research

**This course explains how robot geometry shapes
motion planning, drawing on recent advances in research.**

Serial robots and motion planning

01 Understand the model

Coordinate systems and descriptions of robot geometry

02 Analyse the constraints

Find joint positions that reach a target and respect motion limits

03 Understand global properties

Find which robot postures can connect through continuous motion

04 Use redundancy

Use extra joints to gain more ways to complete a task

05 Verify the complete motion

Check that the robot can follow the entire path within its limits

BY THE END OF PART I

**Plan a continuous motion
and *justify its feasibility.***

Course examples

PUMA 560 robot · 6 joints

Widely used industrial robot

KUKA iiwa LWR robot · 7 joints

Widely used collaborative robot

Custom robots with 3 and 6 joints

For in-depth analysis of robots



Arda Yigit

Tenured researcher
CNRS

Course ENG-654



Postdoctoral researcher

Robotics Laboratory
Parallel robots and physical
human–robot interaction



Tenured researcher

CNRS
Extrinsic actuation
and soft robotics

10 years of experience

Robot design and model-based control



IEEE Transactions on Robotics



Journal of Mechanisms
and Robotics

This course provides you with a survival kit for parallel robots.

Parallel robots: design, kinematics and interaction



The Delta robot

The most famous parallel robot.
Design patented by
Professor Reymond Clavel, EPFL.

Serial robots

One chain of connected joints

Combine the motion of each joint

Motors often move with the arm

Parallel robots

Several chains supporting one platform

Solve for all connected chains together

Motors at the base can reduce moving mass

Design analysis

How geometry shapes the reachable area and performance

Kinematics and singularities

Possible postures and limits on motion and control

Interaction

How easily a person can guide the robot by hand

Redundancy for intuitive interaction



Synthesis and Control of a Parallel Robot for Low-Impedance Sensorless pHRI using Proprioceptive Actuation

Arda Yiğit, Simon Foucault, Thierry Laliberté, David Breton, Zhou Zhou and Clément Gosselin

Laboratoire de Robotique de l'Université Laval
Québec, QC, Canada



Laboratoire de
robotique
de l'Université Laval



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LAVAL

One robot. Two modes.

Redundancy provides extra freedom to adapt the robot's motion during physical interaction.

Handle free

Task mode

Handle grasped

Collaborative mode

Intuitive human–robot interaction.

Stiff control for task execution; easy hand guidance for interaction.

Parallel robots and their design

01 Parallel robot topologies

How links and joints are connected

02 Different types of parallel robots

Compare designs and the motions they allow

03 Kinematic modelling

Relate joint movement to platform movement

04 Singularities and their geometric understanding

Find postures where motion or control is lost

05 Kinematic and actuation redundancy

Extra joints or motors that add design choices

BY THE END OF PART II

**How to design
parallel robots.**



← → ↺ epfl-lasa.github.io/eng-654/ VPN

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ENG ENG-654 SCHEDULE OVERVIEW INSTRUCTORS CONTENT LECTURES EXERCISES EVALUATION RESOURCES COURSE STRUCTURE

EPFL, ROBOTICS, CONTROL AND INTELLIGENT SYSTEMS

Kinematics grounded motion planning for robots

Students learn forward and inverse kinematics, singularity analysis, and constraint-aware path planning for modern robotic manipulators, then implement kinematic models and planners in a final project.

Schedule

Part I is taught on 14 and 16 September. Part II is taught on 22 and 23 September. All courses and exercises are taught in **MED 0 1618**.

COURSE DETAILS

ENG-654

CREDITS	4 ECTS
LANGUAGE	English
ASSESSMENT	50% project report 50% oral presentation

Evaluation

Project teams, assessment weighting, submission deadlines, and report and presentation requirements.

How is the project organised?

The project will be completed in **groups of two students**.

How will the course be evaluated?

The final evaluation is based on the **project report (50%)** and the **oral presentation (50%)**.

When is the project report due?

Submit the project report by **16 November 2026 at 18h00**. A **one-hour grace period** is allowed, until **19h00 on 16 November**.

Submissions after this grace period will be penalized 1 point for lateness.

What are the project report requirements?

When is the oral presentation, and how is it organised?

Oral presentations take place on **25 November 2026**. Each team has **20 minutes: 10 minutes presenting** followed by **10 minutes of oral questions**.

Both students must present for equal time: **5 minutes per student** during the presentation.